Ansh Rathod

+1 (617)319-1525 | rathod.ans@northeastern.edu | linkedin.com/in/anshr21 | github.com/ansh21080 | Portfolio

EDUCATION

Master of Science in Information Systems Sep 2023 – Dec 2025
Northeastern University, Boston, MA
Relevant Coursework: Python & Analytics Technology, Business Processes, Predictive Analytics, Probability Theory & Statistics

Bachelor of Technology in Computer Science Jul 2018 – Jun 2022
National Institute of Technology Nagpur, India
Relevant Coursework: Natural Language Processing, Machine Learning in Python, Big Data, Data Structures & Algorithms

PROFESSIONAL EXPERIENCE

AI/ML Engineer Aug 2025 – Mar 2026
Baxter | Deerfield, IL

- Built a cloud data pipeline using Python, AWS Glue, PySpark, and EC2, cutting execution time from 45 to 10 minutes, and developed a Power BI star schema dashboard reducing report generation by 30%.
- Analyzed 5 years of claims data with Python (SciPy, Pandas), t-tests, and Chi-squared tests to reduce readmission rates by 15%; optimized SQL queries for audit and risk detection, exploring XGBoost and NLP models for predictive insights.

Data Analyst Sep 2024 – Feb 2025
Northeastern University | Boston, MA

- Optimized SQL pipelines in Snowflake to process 15,000+ student applications, cutting query runtime by **30%** and delivering timely admissions insights.
- Automated data categorization using SQL window functions and CTEs; built Power BI dashboards providing real-time applicant tracking for the admissions team.

Software Developer Jun 2022 – Jul 2023
iSoftwareFactory Services LLC | Mumbai, India

- Architected and deployed a full-stack Q&A platform using MongoDB, Express.js, React.js, and Node.js (MERN) with scalable RESTful APIs, reducing production response times.
- Integrated Razorpay payment gateway for subscription-based features; shipped releases on schedule using Agile/Scrum and Git-based version control.

Data Science/Business Analytics Intern Jul 2021 – Dec 2021
Ernst & Young | Mumbai, India

- Developed and deployed a robust XGBoost model with dimensionality reduction for high-cardinality features, achieving 94.9% cross-validation accuracy while reducing computation cost and improving generalization.
- Implemented clustering-based signer identification workflows using DBSCAN, hierarchical clustering, and threshold-based clustering, with automated parameter optimization for improved cross-document robustness.

PROJECTS

Generative AI Supply Chain Analytics Tool Sep 2024 – Dec 2024

- Built a Retrieval-Augmented Generation (RAG) system using OpenAI GPT-4, Pinecone, and Streamlit for semantic search and supply chain decision-making, improving forecasting precision by **25%**.
- Automated data collection and preprocessing using BeautifulSoup, Selenium, Pandas, and NumPy; engineered a flexible pipeline supporting weather data integration.

Voice4Rights — Human Rights Advocacy Platform May 2024 – Jul 2024

- Built a scalable web platform using Node.js, React.js, TypeScript, MongoDB, and Redux with RESTful API-driven donation processing, increasing user engagement by **40%**.
- Deployed an AI-powered chatbot via REST APIs for real-time user support, reducing inbound inquiries by **25%**.

TECHNICAL SKILLS

Programming Languages: Python, R, SQL, MATLAB, JavaScript, Java, HTML/CSS, C

Database & Cloud Tools: MySQL, AWS (S3, EC2, Bedrock, Glue), MS Azure, Git, React, Docker, GCP, Pinecone, Apache Spark, Hadoop, Streamlit, Redshift, Snowflake, Talend, Alteryx, NoSQL, Power Query, Power Automate, Azure Data Factory

Deep Learning: NLP Models, LangChain, OpenAI, PyTorch, TensorFlow, scikit-learn, Hugging Face, Pandas, Keras

Tools & Technologies: Excel (VBA, VLOOKUP), Tableau, Power BI, Oracle SQL, MongoDB, Google Analytics, Adobe Analytics, t-test, Alteryx, Shiny, Streamlit, DBT, Data Warehousing, Data Mining, PowerPoint, MS Office, Looker

AI & Generative AI: LangChain, OpenAI API, Claude (Anthropic), Hugging Face, Retrieval-Augmented Generation (RAG), LLM Fine-Tuning, Prompt Engineering, AI Agents, Vector Databases (Pinecone), AWS Bedrock, Multimodal AI